

GIACOMO MARCIANI

SENIOR SOFTWARE ENGINEER @ Amazon Web Services
Building High Performance Computing in the Cloud



LOCATION

Boston, MA, United States

CONTACTS

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INTRODUCTION

Senior Software Engineer with 8+ years at AWS, specializing in High Performance Computing, distributed systems, cloud infrastructure, infrastructure automation, and technical leadership. I own the full lifecycle end-to-end: strategy, system design, implementation, CI/CD, observability, security, and on-call operations.

I have contributed to AWS ParallelCluster, NICE EnginFrame, Elastic Graphics, and Elastic Inference. My background spans software engineering, distributed systems and big data. Before AWS, I founded the first ACM Student Chapter in Rome, co-founded a startup, and taught computer science at university. I also led research projects that resulted in publications in scientific journals and presentations at international conferences.

What makes me effective as an engineer and a leader is a relentless curiosity; a drive to keep improving how I build, how I lead, and how I show up for my team.

WORK EXPERIENCE

Sr Software Engineer

Amazon Web Services, 2021-Present

Technical leader of AWS ParallelCluster. My responsibilities span the software lifecycle end-to-end: strategy, design, implementation, observability, on-call incident response, mentoring, and hiring.

Software Engineer II

Amazon Web Services, 2019-2021

Worked on Elastic Graphics, Elastic Inference, and NICE EnginFrame. On EnginFrame, I contributed mainly to the build system, CI/CD automation, and integration work to enable the portal to interact with both on-premises and cloud-based clusters. Starting in 2021, I took on AWS ParallelCluster, where I became the team's de facto technical leader. My key contributions include Multi-User environment, Dynamic File System Mounting, support for GPU-intensive instance types, enabling ParallelCluster in US isolated regions, Terraform integration, ParallelCluster UI, serving as Security Guardian, driving roadmap and releases, and resolving high-severity escalations for key customers. I also led the complex shutdown of Elastic Graphics and Elastic Inference, navigating deep integrations with EC2/ECS.

Software Engineer I

Amazon Web Services, 2018-2019

Worked on Elastic Graphics and Elastic Inference, which attached cost-effective GPU acceleration to EC2 instances and other AWS services. I contributed to the launch of Elastic Inference at Re:Invent 2018 and owned the metering and billing component for compute resources.

R&D Intern

Accenture, 2017

Designed a solution for the elastic auto-scaling of containerized applications using Reinforcement Learning in Kubernetes.

Chairman, Founder

ACM Rome Tor Vergata, 2016-2018

Founded the ACM Student Chapter at University of Rome Tor Vergata, the first in the city. Led the team and fostered connections between academia and industry.

Teaching Assistant

University of Rome Tor Vergata, 2016-2018

Tutored undergraduate students and graded exams for Foundations of Computer Science and Algorithms Engineering.

Mathematics Tutor

Freelance, 2011-2017

Gave private math lessons to secondary and high school students. Several of my students went on to study engineering.

Management Engineer, Co-Founder

Sbaam.com, e-Commerce, 2011-2012

Co-founded the startup, making it a finalist in the InnovActionLab startups contest. Responsible for business plan development and investor pitches.

MAIN PROJECTS

AWS ParallelCluster and ParallelCluster UI

Amazon Web Services, 2021-Present | HPC

Open-source platform to provision and operate large-scale HPC clusters on AWS, automating compute, scheduling, storage, and networking. Drives multi-million-dollar revenue and powers demanding workloads for major enterprise and research customers.

NICE EnginFrame

Amazon Web Services, 2021 | HPC

Web portal to submit and manage HPC workloads across on-premises and AWS clusters, offering a unified interface for job submission, monitoring, and remote 3D visualization.

AWS Elastic Graphics and Elastic Inference

Amazon Web Services, 2018-2023 | Web Service

Services attaching low-cost GPU acceleration for OpenGL and ML inference to EC2, ECS, and SageMaker, cutting the cost of running graphics and machine learning workloads at scale.

Crime Graph

Big Data, IEEE FiCloud, 2017 | Java, Kafka, Flink, Neo4j

Data-stream processing application for real-time analysis of evolving criminal networks, applying graph algorithms over Kafka and Flink to surface hidden and latent relationships.

CLI Wizard

Developer Tools, 2025 | Python, Jinja

Code generator that produces complete, pip-installable Python CLI projects from OpenAPI v3 specs, with automatic command grouping, profile management, and YAML-based customization.

PyDES

Performance Modeling, 2019 | Python

Discrete-event simulation suite for performance modeling of cloud systems, featuring batch-means analysis, Markov-chain solvers, and multi-stream pseudo-random number generation.

OntoQA

Artificial Intelligence, 2017 | NLP, Java, Protege

Question-and-answer web application leveraging ontology-driven natural language processing to map free-text questions onto a knowledge graph and return precise answers.

Godot

App for Transports, 2013 | Java, Android

Mobile app that tackles illegal parking by letting citizens report violations through QR codes and NFC tags, streamlining the flow of reports to local transport authorities.

See more on my GitHub

PUBLICATIONS

A data streaming approach to link mining in criminal networks

5th IEEE International Conference on Future IoT (FiCloud 2017)

Real-time analysis of social networks leveraging Flink

10th ACM International Conference on Distributed and Event-Based Systems (DEBS 2016)

Marciani Normal Form of context-free grammars

Arxiv (2016)

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EDUCATION

MSc Software Engineering and Data Science
University of Rome Tor Vergata, 2015-2018
Grad Avg 29/30 - GPA 4/4 *

BSc Software Engineering
University of Rome Tor Vergata, 2011-2015
Grad 110/110 cum laude

BSc Management Engineering
University of Rome Tor Vergata, 2009-2011
Interrupted

MAIN SKILLS

- Cloud** Amazon Web Services, Digital Ocean, Railway
- HPC** ParallelCluster, EnginFrame, Slurm, Nvidia GPUs, Cuda, MPI, Lustre, EFA
- Big Data** Flink, Hadoop Distributed File System, Hadoop MapReduce, Pig, Hive, Kafka, Flume, Elasticsearch, Kibana, Grafana, Kubernetes
- Web** Spring, Nginx, Tomcat, React, NodeJS, HTML, CSS, Sass, Postman, Hugo, Wordpress
- Automation** CloudFormation, Terraform, Chef, CI/CD, Infrastructure as Code
- Observability** CloudWatch, Grafana, Kibana, Logging, Metrics, Alarms
- Security** Threat Modeling, Security Reviews, IAM, Encryption, Compliance
- Data** SQL, NoSQL, Graph, Caching
- Languages** Python (primary), Java, Kotlin, Ruby, Javascript, C, Cuda, Bash, HTML, OWL, SPARQL, AMPL, MIPS, Latex
- Platforms** Linux, OSX, Docker
- Applied Math** Matlab, AMPL, IBM Cplex
- Leadership** Technical Leadership, System Design, Mentoring, Hiring, Cost Optimization
- Process** Agile, SCRUM

HONOURS

- Lecturer at Master CESMA**
University of Rome Tor Vergata, 2016
- Finalist at ACM DEBS Grand Challenge**
10th ACM International Conference on Distributed Event Based Systems, 2016
- Speaker at InnovAction Lab Grand Challenge**
LVenture Groups, 2013
- Best Student in Computer Architectures**
University of Rome Tor Vergata, 2012
- Alumno of InnovAction Lab**
LVenture Groups, 2011
- Counselor of Engineering**
University of Rome Tor Vergata, 2010

LANGUAGES

Italian
Native

English
Proficient